



Glossary

5K:

Running event/race (3.1 miles)

10K:

Running event/race (6.2 miles)

Abduction:

Anatomical directional term meaning moving away from the body.

Acetabulum:

The location of the pelvis that the head of the femur sits in.

Acetyl-CoA:

A molecule that is the result of oxidation of fatty acid, amino acids and pyruvate. Acetyl-CoA is broken down and used for energy production in the Krebs Cycle.

ACSM:

American College of Sports Medicine

Acromion Process:

Bony aspect of the shoulder blade that extends above the shoulder joint. It is joined to the clavicle via the *Acromioclavicular Joint* (AC Joint).

Adduction:

Anatomical directional term meaning moving toward the body.

Adenosine Triphosphate (ATP):

A molecule that is responsible for storing and releasing energy in the body. Due to this, biologists have coined ATP the “Currency of Life”. ATP is generated in the mitochondria of cells. ATP can be produced aerobically and anaerobically. The body stores only a small amount of ATP at any given time. Therefore ATP must be continually regenerated on a cellular level.

AED:

Automated External Defibrillator. Portable device that is used to diagnose heart rhythms and provides an electronic shock if needed to re-establish a normal heart rhythm.

Aerobic:

Requiring oxygen

Agonist:

Muscle or muscles that are primarily responsible for movement around a joint. Also called a “prime mover”.

Alveoli:

Air sacks in the lungs that represent the end of the respiratory tract.

Anaerobic:

This term means “without oxygen”. This is a slight misnomer as being in an anaerobic state typically has more to do with a lack of utilization of oxygen rather than it not being present.

Antagonist:

Muscle or muscles that act in opposition to the *Agonist* muscle(s)

Anterior:

Anatomical directional term referring to the front or toward the front of the body.

Anterior Superior Iliac Spine (ASIS):

Furthest anterior part of pelvis (bilateral). Commonly used as a landmark for assessments.

Anthropometric:

A measurement of a site or place on the human body.

Appendicular Skeleton:

Consisting of 126 bones, these bones are responsible for locomotion.

Arteries:

Carries oxygenated blood away from the heart to the body.

Asthma:

A chronic inflammatory response of the airways.

Atrophy:

Reduction in muscle size. Often caused by prolonged inactivity.

Autonomic Nervous System (ANS):

Controls cardiac and smooth muscle contractions and does not require conscious thought like the central nervous system. The ANS consists of two nervous systems, sympathetic and parasympathetic that serve to self-regulate the body.

Axial Skeleton:

Consisting of 80 bones, the axial skeleton provides the central structure of the body. Major parts of the axial skeleton include the skull, spine, ribs and sternum.

Ballistic Stretch:

A type of passive stretch that involves “bouncing” or rapid movements. This type of stretching is not recommended due to the high potential for injury.

Base Layer:

Type of athletic clothing that is worn right next to the skin for the purpose of wicking sweat and keeping the skin dry.

Base Training:

Training practice typically done for the purpose of increasing aerobic fitness and endurance. Characterized by low to medium intensity and at a steady state pace.

Biceps Brachii:

Two-headed arm muscle that flexes the elbow. The muscle originates at the scapula and humerus and inserts on the forearm (radius/ulna).

Bilateral:

Pertains to movement by both sides of the body.

Bioelectric Impedance Assessment (BIA):

Type of body composition measurement that uses a low level electrical impulse. Based on the resistance to the flow of the electricity, the body composition can be derived.

Biomechanics:

The study of the mechanics of the human body

Blood Lactate Analyzer:

A device that analyzes a small blood sample to determine the level of lactate within the blood. Used to determine training effort level parameters.

Body Mass Index (BMI):

BMI is calculated using a formula that attempts to determine if an individual is underweight, average weight, or overweight. This is determined by comparing an individual's height and weight. BMI is not an accurate tool to use for an athlete as it does not take body fat or muscle into consideration.

Brachioradialis:

Forearm muscle that extends the wrist.

Burn Out:

A mental state at which an individual has lost interest in an activity and often coincides with exhaustion.

Bursitis:

Inflammation of a bursa sac. A bursa sac is a synovial fluid-filled sac that is located between bone and tendon, or tendon and skin. The purpose of a bursa sac is to provide cushioning and reduce friction.

Cadence:

Used to describe the foot turnover at which an individual runs. Typically denoted in, steps per minute.

Calves:

Muscles behind the bone (tibia) of the lower leg. Consisting of the gastrocnemius and soleus.

Capillaries:

The smallest blood vessels in the body. They increase in density based on how much blood supply is needed in a particular area. They are found in the tissues of the human body and transports blood from the arteries to the veins. Due to the thinness of the walls of capillaries, O₂, CO₂, waste products and nutrients can pass through the walls.

Carbohydrates:

Consisting primarily of sugars and starches. Carbohydrates make up one of the three principal types of nutrients used as energy by the body. Carbohydrates are often classified as *Simple* and *Complex*. The classification depends on the chemical structure of the food, and how rapidly the sugar is digested and absorbed by the body.

Carbohydrate Loading:

Characterized by eating additional carbohydrates in the days leading up to an event with an associated decrease in training volume. The goal of carb loading is to maximize endurance performance by increasing the muscle's glycogen levels. The greater the glycogen stores, the longer aerobic activity can last.

Cardiac Drift:

A phenomenon that relates to a natural increase or upward "drift" in heart rate while the overall intensity (i.e. respiration rate, effort level, caloric burn) stays the same.

Cardiac Muscle:

Located only in the heart and is solely responsible for contracting/relaxing the heart to pump blood to the lungs and the rest of the body

Cardiac Output:

The volume of blood being pumped out of the heart in one minute.

Cartilage:

Cartilage is primarily made up of collagen and water and is not supplied by blood or nerves. There are three types of cartilage; *Hyaline (Articular)*, *Fibrocartilage* and *Elastic*. In regard to movement, the most important to be aware of is Hyaline cartilage. Hyaline cartilage is found on the surfaces of most joints. Hyaline cartilage protects the joints and acts as a barrier between bones.

Central Nervous System (CNS):

The CNS is a voluntary nervous system, meaning it requires conscious thought to be activated. The brain sends a signal to the motor neurons that innervates muscles fibers, thus producing skeletal muscle contractions.

Cervical Spine:

The highest part of the spine. The cervical spine is made up of 7 vertebrae, C1-C7. The higher the number, the closer to the head the vertebrae is.

Circumduction:

The combination of abduction, adduction, flexion and extension. Moving a limb in a circular motion represents a circumduction motion.

Clavicle:

Often referred to as the collarbone. Connects the sternum and scapula.

Client Profile:

The purpose of the client profile is to get a wide variety of information to better and more accurately assess/coach them.

Closed Chain Exercise:

Exercise where the hand or foot is fixed and cannot move.

Coccyx:

Commonly referred to as the “tailbone”, the coccyx is made up of between 3-5 fused bones and represents the base of the vertebral column.

Compression Socks:

Socks that extend most of the way up the calf and “compress” the calf muscles. Thought to aid in reduction of muscle vibration, increase blood flow back to the heart and muscle recovery.

Concentric:

Shortening of a muscle that produces force.

Contralateral:

On the opposite side of another structure. Lifting the right arm is contralateral to the left leg.

Core Muscles:

Muscles that are involved with stabilizing and providing support to the pelvis and spine. Core muscles include: Rectus abdominis, external obliques, internal obliques, transverse abdominis, erector spinae, hip flexors (psoas) and gluteal muscles.

Cori Cycle:

The metabolic pathway in which lactate is produced by anaerobic glycolysis in the muscles, moves to the liver and is converted to glucose and then returns back to the muscles to be converted back to lactate.

CPR:

Cardio Pulmonary Resuscitation. The purpose of CPR is to provide partial blood and oxygen flow to the heart and brain.

Creatine Phosphate (CP):

A high-energy phosphate compound that aids in the re-synthesis of ATP.

Cross Training:

Any exercise that is done for the purpose of training that does not consist of running.

Deep:

Directional term used to describe a structure that is away from the surface of an individual.

Dehydration:

Caused by an individual not taking in fluids at the same rate as it is expending them. Dehydration can range from mild to very severe in which death can result.

Delayed Onset Muscle Soreness (DOMS):

Muscle soreness that reaches its peak soreness 24-48 hours after the exercise bout ended. Shown to primarily result from eccentric related movements.

Deltoids:

Shoulder muscles. Typically not used to describe the muscles of the rotator cuff.

Depression:

Downward motion of body part

Desensitization:

The purpose of desensitization is to reduce and/or eliminate phobias through repeated and increased exposure to the phobia.

Detraining:

Periods of substantially reduced activity or a lack of physical activity. Correlates with a decrease in fitness level.

Diaphragm:

The diaphragm is a “dome-like” sheet of muscle located between the thoracic and abdominal cavities and is responsible for the increase and decrease of lung size when inhaling (contraction of diaphragm) and exhaling (relaxation of diaphragm), respectively. The diaphragm also provides lumbar stability by working with the other *Inner Unit* core muscles.

Distal:

Away from the point of attachment, further from the center of the body. The tibia is distal to the femur.

Dorsi Flexion:

Decrease angle between lower leg and foot. Walking on your heels results in the feet being dorsi flexed.

Dynamic Stretch:

A stretch where movement is involved to facilitate the stretch.

Eccentric Contraction:

Lengthening of a muscle and typically used as a “braking” force to counteract the concentric contraction.

Electrolytes:

Minerals that break into small electrically charged particles called ions. Present wherever there's water in your body, electrolytes regulate the body's fluids, helping to maintain a healthy blood pH balance, and creating the electrical impulses essential to all aspects of physical activity - from basic cell function to complex neuromuscular interactions needed for athletic performance. The kidneys work to keep the electrolyte concentrations in the blood constant despite changes in the body. One of the many important electrolytes is sodium.

Electron Transport Chain:

Takes place in the membrane of mitochondria and is the last phase of the oxidative phase. It is responsible for the majority of ATP generation.

Elevation:

Upward motion of a body part(s)

Excess Post-Exercise Oxygen Consumption (EPOC):

Oxygen intake after exercises ceases. The purpose of EPOC is to eliminate oxygen debt.

Erector Spinae:

Also called, *Extensor Spinae*, this group of muscles and tendons run parallel to the vertebral column from its base (lumbar) through the top (cervical) section. The muscles that make up the erector spinae are: *Illiocostalis* (Lateral), *Longissimus* (Center), *Spinalis* (Medial). The primary function is extension and lateral flexion of the spine. Secondly, it provides isometric stabilization to the spine.

Eversion:

Turning outward. When the sole of the foot is facing outwards, the foot is everted.

Extension:

Increase of a joint angle

External Abdominal Oblique:

This muscle is located to either side of the rectus abdominis and is responsible for flexion of the spine as well as rotational and lateral flexion (bending side-to-side) of the upper body. Like the rectus abdominis, it secondarily assists in stabilizing the lumbar spine via isometric contraction.

Fartlek:

From the Swedish word: *Speed Play*. Used by runners to describe a workout with changes in tempo, typically unstructured.

Fascia:

A 3D web of connective tissue that surrounds all muscles, bones, nerves and blood vessels. Has a large affect on posture and muscle balance. Restricted fascia creates pain and muscle dysfunction.

Fast Twitch Muscle Fiber:

Often referred to as "Type II" fibers, fast twitch fibers are broken up into two types: IIA and IIB. The primary characteristic of fast twitch muscle fibers is that they contract with a high degree of force but lack the ability to fire repeatedly without fatigue. Fast twitch fibers are non-oxidative, meaning they do not require oxygen to contract.

Fat:

A major energy source for the body during long endurance bouts. The primary location for fat storage is adipose tissue (i.e. body fat) within the body. The majority of the adipose tissue that is used for energy is called subcutaneous fat (below the skin). Adipose tissue is comprised of *Adipocytes*, or fat cells. Adipocytes are where triglycerides are stored and synthesized.

Femur:

The only bone in the thigh. Connects the pelvis to the tibia/knee.

Flat Feet:

Flat feet are often called “fallen” or “collapsed” arches. Individuals with flat feet have little to no arch in the feet in the standing position. Those that overpronate often have “flat feet” but this is not always the case.

Flexion:

Decrease of a joint angle

Float Run:

Essentially *Repeats* but instead of an easy jog in between efforts, the pace still relatively high. Float workouts train the body to be able to respond to mid-race surges while not dipping below race pace.

Foam Roller:

Essentially a semi-hard, round tube of Styrofoam. Foam rollers come in varying degrees of hardness as well as lengths. The idea behind foam rollers is that they are placed under a tight muscle or area of connective tissue and the body “rolls” over an area of tightness until the area becomes less restrictive and tight.

Force:

A pushing or pulling motion between two objects. The product of mass and acceleration.

Foundation Exercises:

A series of resistance exercises that together provide a solid strength base for endurance athletes

Fracture:

A break in a bone. Often caused by impact or repeated stress.

Friction:

A force acting on the area of contact between two surfaces in the direction opposite the motion or motion tendency.

Frontal Plane:

Divides the body into front and back halves. Stepping to the side is a movement in the frontal plane

Fulcrum:

Point of support on which a lever pivots.

Functional Training:

Resistance training exercises that mimic the range of motion of the activity being trained for.

Gastrocnemius:

Superficial muscle, located on top of soleus and behind the tibia. Acts to plantar flex and stabilize the ankle.

General Adaptation Syndrome (GAS):

Developed by Hans Selye, the GAS describes how individuals adapt to stress and consists of three phases: Alarm, Resistance, and Exhaustion. Adapting GAS to physical training, it demonstrates that the body can get stronger though adapting to stress but once the stress gets too high, the body will begin to break.

Girth Measurement:

Measurement (circumference) of a body part

Glenohumeral Joint:

Shoulder joint

Glucose:

Blood sugar. A energy source for the body's cells

Gluteal Muscles:

Comprised of the *gluteus maximus*, *gluteus medius* and *gluteus minimus*. The primary role of the gluteal muscles is to support the LPHC and to extend and externally rotate the hip.

Glycogen:

Main form of carbohydrate storage in the body (muscles/liver) and can be converted to glucose.

Glycogen Depletion:

Lack of glycogen in the body often results in the loss of energy and extreme fatigue.

Glycolysis:

Metabolism of glucose

Golgi Tendon Organ (GTO):

The GTO is a nerve receptor that is located at the junction of the muscle and tendon. During a stretch, the GTO senses tension of the muscle/tendon and sends a signal to the spinal cord. The spinal cord then sends a signal back to the muscle to relax. This is called the *Golgi Tendon Reflex*. In other words, the GTO exists to regulate the amount of tension on a muscle or muscles so that injury does not occur to the muscle(s) and connective tissue.

Goniometer:

A measurement tool to assess angles of the body.

Greater Trochanter:

Lateral protrusion at the top of the femur

Guerilla Marketing:

A form of business marketing which typically is very impactful and free or very low cost.

Gulick Tape Measure:

A special tape measure used to measure body girth measurement. The tape incorporates an expanding spring so that the tape cannot be over tightened, Such over tightening would result in a smaller than actual girth measurement reading.

Half-Marathon:

Running event/race (distance of 13.1 miles)

Hamstrings:

The hamstrings are comprised of three muscles:

- Biceps Femoris
- Semitendinosis
- Semimembranosus

These muscles are used to stabilize the LPHC, flex the lower leg and extend (posterior) the femur

Health History Questionnaire (HHQ):

Enables a coach to get a thorough background of their client that would be pertinent to the physical training process.

Heart Rate Monitor (HRM):

Typically worn as a watch, an HRM reads in real-time an individuals heart rate.

Herniated Disk:

Caused when excessive stress placed on an intertebral disc causes the annulus fibrosis to crack and some of the nucleus pulposus leaks out.

High Intensity Intermittent Training (HIIT):

Characterized by an intensity level that is equal to or greater than that of intervals. While the time frame for bouts of intensity/recovery might be close to that of intervals, HIIT often incorporates varying modalities of training (i.e. Rowing, strength training, plyometrics, etc...)

Hip Adductors:

There are five primary muscles that are responsible for hip adduction.

1. Pectineus

2. Adductor Longus
3. Gracilis
4. Adductor Magnus
5. Adductor Brevis

Hip Flexors:

Muscles that flex the hip. Primary hip flexors are the psoas (major and minor) and iliacus muscles that are often referred to as a muscle group called the *iliopsoas*. The psoas major or iliacus also adduct the hip.

Hyperoxic:

In regard to training, this refers to exercising with supplemental oxygen so that an individual is breathing air more heavily concentrated with oxygen than the environment would allow.

Hyperplasia:

Increase in the number of cells

Hypertrophy:

In terms of muscle development, hypertrophy is an increase in the size of muscle cells, thus increasing the size of a muscle.

Hyponatremia:

A metabolic condition where there is insufficient sodium (i.e. electrolyte imbalance) within the body fluids. This is often the cause of extreme over hydration.

Hypothermia:

A condition where the core body temperature drops below what is required for normal body functioning. This is caused when the body loses heat faster than it can produce heat. Can lead to fatality.

Hypoxic:

In regard to training, this refers to training in an environment that has reduced oxygen levels. This can be done naturally (training at altitude) or artificially (altitude chamber). The goal of hypoxic training is to increase one's red blood cell levels.

Iliac Crest:

Most superior aspect of the pelvis (Ilium)

Iliopsoas:

A group of muscles commonly termed *hip flexors*, due to their primary role. The iliopsoas consists of three muscles: *Psoas (major/minor)*, *Iliacus*.

Illiotalibial (IT) Band Syndrome:

This typically causes pain on the lateral aspect of the knee. The primary cause is the IT Band rubbing against the lateral epicondyle (protrusion at base of femur) repeatedly to the extent that inflammation and pain occurs.

Ilium:

Located on either side of the sacrum, the Ilium is the largest bone(s) in the pelvis. The primary purpose of the Ilium is to protect internal organs.

Indoor Resistance Cycling Trainer:

Often referred to as a *trainer*, this is a unit that attaches to a bicycle's rear wheel. This unit provides resistance to allow an individual to replicate riding outdoors.

Inferior:

Used to describe anatomical direction, "below" or "lower". An example of this would be, the hips are inferior to the chest.

Inner Unit Core Muscles:

Provides stabilization and rigidity to spine and pelvic region. Made up of the *Multifidus*, *Transverse Abdominis*, *Diaphragm*, *Internal Oblique* and *Pelvic Floor Muscles*.

Innervate:

To supply an area (body part) with nerves. To stimulate.

Internal Oblique:

This muscle can be considered the intermediate muscle of the core in regard to its placement, as it lies between the transverse abdominis and the external oblique. In addition to providing spinal stability, it has two additional functions. First, the contraction/relaxation of the internal oblique results in movement of the diaphragm, which is responsible for inhalation and exhalation. Second, the internal oblique is responsible for rotation and lateral flexion (side-to-side bending) of the upper body.

Intervals:

A training method that is characterized by structured time periods of intense exercise followed by recovery periods.

Intervertebral Disks:

These disks are located in between spinal vertebrae and are made up of fibrocartilage. They allow the spine to move and absorb compression.

Inversion:

Turning inward. When the sole of the foot is facing inwards, the foot is inverted.

Involuntary Muscle Contraction:

This type of muscle contraction is controlled by the autonomic nervous system (ANS) and occurs without conscious thought. An example of this is the contraction of the heart.

Ipsilateral:

On the same side as another structure. Lifting the right arm is ipsilateral to the right leg.

Isokinetic:

Muscle contraction where the shortening/lengthening of the muscle occurs at a set speed. The speed of the muscle shortening/lengthening is usually controlled by an outside source.

Isometric:

Type of muscle contraction characterized by a lack of movement. Therefore the muscle length stays the same but has tension on it. Holding a 15 lb. dumbbell with the elbows at 90 degrees is an example of an isometric contraction.

Isotonic:

Type of muscle contraction characterized by the muscle shortening with a constant load applied to it. A bicep muscle performing a curl holding a 15 lb. weight is an example of an isotonic contraction.

Illiotalibial (IT) Band:

A long fibrous band that originates on the iliac crest of the pelvis and attaches on the lateral condyle of the tibia. It is responsible for stabilization of the knee. The glute maximus and tensor fascia latae insert into the IT band.

Kinematics:

Describes the appearance of motion

Kinesiology:

The study of human movement

Kinetic Linking:

Relates to the order in which muscles activate and work together to allow the body to move. From a performance-based standpoint, it also pertains to how muscles work to give an individual a biomechanical advantage. Also referred to as the *Kinetic Chain*.

Kinetics:

Studies the forces associated with motion

Krebs Cycle:

A cycle of complex chemical reactions that cells go through in the presence of oxygen to create energy (cellular respiration).

Kyphosis:

Excessive thoracic spinal curvature (hyperflexion)

Labrum:

Found around the edge of the shoulder and hip joints. Its primary purpose is to keep the ball aspect of the humerus and femur within the shoulder and hip socket, respectively. It is comprised of fibrocartilage. Injury to the labrum is called a *Labral Tear*.

Lactate:

An enzyme found in body tissue. During high intensity efforts where oxygen supply is limited, pyruvate produces lactate which allows for the break down of glucose for energy. Lactate also helps to buffer cellular acidosis.

Lactate Clearance:

The process by which the body utilizes lactate. Lactate be “cleared” in one of two ways:

1. Lactate can be cleared (i.e. utilized) oxidatively within the muscle cell that it was produced in, or it can be transported to other muscle cells to be cleared.
2. Lactate that is not oxidized can be cleared anaerobically via the Cori Cycle. This occurs in the liver.

Lactate Threshold:

Indicative of the level at which blood lactate begins to accumulate in the blood stream. This occurs at the point at which the rate of lactate production increases faster than the rate of removal (clearance).

Lateral:

Anatomical directional term referring to moving away from the body.

Lateral Femoral Epicondyle:

A small, lateral, protruding aspect at the inferior aspect of the femur. Often used as a landmark, it is located to the outside of the knee. This landmark is often referenced as the main source of pain from *Iliotibial Band Syndrome* (IT Band Syndrome), as the epicondyle rubs against the IT Band when running and cycling.

Lateral Rotation (external rotation):

Rotation away from the midline of the body. Someone who is “bow-legged” has lateral rotation of the upper leg (femur).

Latissimus Dorsi:

Often referred to as the “lats”, the latissimus dorsi is the largest muscle of the back. Its primary functions are to adduct the arms, medial rotation of the humerus and downward rotation of the scapula. It also assists in stabilizing the LPHC.

Latitudinal Axis:

Vertical axis (often referred to as the “Y” axis). Spinning around on one leg while standing represents rotation around the latitudinal axis.

Leg Length Discrepancy (LLD):

One leg is longer than the other. There are two classifications of leg length discrepancies. A *Functional LLD* means one leg appears longer than another due to injury or muscle imbalances that cause the hips to tilt. An *Actual LLD* means that one leg is structurally longer than the other.

Length/Tension Relationship:

If a muscle is shortened or lengthened too much, force production of the muscle will diminish to the point where minimal or no force production is possible. The relationship between a muscle’s length and the ability to produce force is a bell curve.

Levator Scapulae:

Assists in cervical extension. This muscle is located at the base and back of the neck.

Ligament:

A fibrous tissue that connects bone to bone.

Long Muscle:

Muscle is too elongated (i.e. stretched) to provide adequate force production. This is often termed an *inhibited* muscle.

Long Steady Distance (LSD):

A type of endurance sport workout that corresponds to training over a long distance, and at a relatively easy to moderate, steady pace.

Lordosis:

Excessive lumbar/cervical spinal curvature (hyperextension)

Lumbar Spine:

Often referred to as the low back, the lumbar spine consists of 5 vertebrae (L1-L5) and is located between the thoracic aspect of the spine and the sacrum.

Lumbo-Pelvic-Hip Complex (LPHC):

Commonly referred to as the *Core*. The LPHC specifically addresses the area that stabilizes the upper and lower body. Therefore, muscles that act to stabilize this complex can be viewed as core musculature.

Marathon:

Running event/race (distance of 26.2 miles)

Maximum Lactate Steady State (MLSS):

The highest intensity level at which blood lactate concentrations are maintained at a steady state (equilibrium) level during exercise bouts of approximately 60 minutes.

Medial:

Anatomical directional term relating to being closer to the midline of the body.

Medial Rotation (internal rotation):

Rotation toward the midline of the body. Someone who is “knock kneed” has *medial rotation* of the upper leg (femur).

Medicine Ball:

A round ball that comes in varying weights and sizes and used for the purpose of strength and conditioning.

Mental Imagery:

A technique used by athletes to enhance performance. External imagery occurs when an individual “sees” him or herself performing an activity with the goal of visualizing the activity performed perfectly. Internal imagery occurs when an individual “feels” how they anticipate feeling during the event to prepare hi or herself for the event. Internal imagery can also relate to visualizing from a first person point of view.

Midtarsal (a.k.a. *Transverse Tarsal* and *Talocalcaneonavicular*):

This joint spans across the arch of the foot and primarily controls inversion and eversion of the foot. This joint ranges from flexible to rigid based on what aspect of the support and drive phase the foot is in. When running or walking, as the foot moves under the body’s center of gravity and prepares to push-off, the joint locks and becomes rigid to provide a stable base of support to push off of.

Minimalist Running Shoe:

A shoe that has minimal heel padding/arch support and often are designed to be lightweight as well. This typically results in a low drop from the heel to the toe. Minimalist shoes are typically designed for running with a midfoot strike.

Mitochondria:

An organelle that is the power plant of cells. Converts the potential energy from food molecules into ATP.

Motor Unit:

A motor neuron and all the muscle fibers it innervates.

Multifidus:

The multifidus (mull-tiff-a-duss) is actually a series of small muscles that attach laterally to both sides of the spinal column and is deep to the erector spinae. The multifidus can be broken down into superficial and deep muscles. The superficial multifidus is responsible for lumbar back extension, rotation and lateral flexion and the deep multifidus is responsible for spinal stabilization. The multifidus co-contracts with the TVA to provide spinal stability.

Muscle Cramp:

Characterized by a muscle involuntarily contracting (i.e. spasms) that is often painful due to the intensity and duration of contraction. Cramping is often caused by glycogen depletion, dehydration, low levels of potassium, exercising for a longer period of time or at a higher intensity than the body is used to.

Muscle Imbalance:

Exists when there is a lack of balance between two opposing muscles that surround a joint. Muscle imbalances can be caused by many things including habitual and repetitive movements, incorrect posture, restricted fascia, injuries or surgeries.

Muscle Spindle:

Muscle spindles are located deep in a muscle's belly. Muscle spindles sense change in muscle length as well as the range of change in length and sends a signal to the spine. This signal triggers the stretch reflex (also called: myotatic reflex). The stretch reflex looks to resist the increase in muscle length by causing the muscle to contract (365). The more rapid the increase in muscle length is, the stronger the stretch reflex is. Therefore, the primary function of muscle spindles is to protect the body from injury due to overstretching.

Muscle Sprain:

Injury to a ligament (ligament connects bone to bone). Sprains are categorized from Level 1-3 with 3 being the most severe

Muscle Strain:

Injury to muscle or tendon (tendon connects bone to muscle)

Muscular Compensation:

Occurs when a muscle is unable to provide some or all of the required muscle activation. This leads to other muscles being activated to assist in performing a body function (i.e. movement of a body part).

Myofascial Release:

Pertains to the reduction of tightness and restrictiveness of the muscles and fascia.

Myofibril:

Contractile fibers that run along the length of a muscle fiber

Myostatin:

Protein that determines the size of a muscle

Negative Splits:

Refers to a structured training or racing strategy that has an individual running the second half of an event or training session at a faster pace than the first half. Negative splits can also refer to performing *Repeats*, with each effort done at a faster pace than the one before it.

Nerve Impingement:

Often called a *Pinched Nerve*. Occurs when a nerve has pressure to it. This pressure often results in pain, numbness or muscle weakness in the area that nerve(s) innervates.

Neuroendocrine Response:

Relates to the interactions between the nervous and endocrine systems that occur to the body when performing specific types of strength training movements. The highest neuroendocrine response occurs when performing exercises that recruit large muscle masses and are done at a high volume with moderate to high resistance. This response increases testosterone and hormones (i.e. growth hormone) that stimulate muscle growth/strength.

Neutral Spine:

Spinal position that requires minimal neuromuscular activity to maintain a relaxed posture in a standing position.

Onset of Blood Lactate Accumulation (OBLA):

Represented when blood lactate levels reach 4 millimoles per liter (mmol/L) during exercise bouts.

Open Chain Exercise:

Exercise where the hand or foot is free to move in the air.

Outer Unit Core Musculature:

The outer core musculature's primary purpose is to produce movement and its secondary purpose is to assist in providing stabilization to the LPHC. Comprised of the *Rectus Abdominis*, *External Abdominal Oblique*, *Erector Spinae*, *Quadratus Lumborum*, *Iliopsoas*, *Hip Adductors*, *Hamstrings*, *Rectus Femoris*, *Gluteus Maximus*.

Overtraining:

The result of too much training stimuli without the proper amount of rest. As a result, the individual cannot recovery properly and thus performance diminishes and in some circumstances can lead to injury.

Overhead Squat Screen (OSS):

One of the best all-around functional tests to assess strength, flexibility and neuromuscular efficiency. The test is administered by having an individual squat down with their weight over their heels and their arms over their head. The form is assessed for postural deviations.

Overload Principle:

A greater than normal stress on the body needs to be present for a training adaptation to occur.

Over Distance Training:

Training a longer distance than the actual race distance

Over Pronation (ankle):

While there is not a clear consensus on exactly how many degrees equates to over pronation, many clinicians believe that a subtalar joint angle greater than 8-10 degrees equates to over pronation.

Overtraining:

The point at which an athlete trains too much and experiences a decline in performance and progress.

Oxidative Phosphorylation:

A metabolic pathway that consists of a series of chemical reactions within the cells, and is the last phase of the aerobic pathway. This phase is responsible for the majority of ATP generation.

Oxygen Debt:

The difference between oxygen consumption at rest and elevated rate of oxygen consumption following an exercise bout.

Pacing:

Relates to a structured plan of intensity and/or speed over the course of a training session or race. This plan is part of the overall strategy for a training session or race.

Patellofemoral Syndrome (PFS):

Characterized by pain behind the kneecap. Commonly referred to as *Runner's Knee*. There are many potential causes of PFS including muscle imbalances of the quadriceps muscle, asymmetrical hip rotation and compressive forces.

Peaking:

A systematic process by which an individual realizes their highest possible physical level of preparedness on a set date or time frame.

Pectoralis Major:

The primary muscle of the chest. It is a fan shaped muscle that originates from the sternum and inserts on the humerus. The primary action is to flex, adduct and internally rotate the humerus.

Pectoralis Minor:

Deep chest muscle. Acts to move the scapula forward and downwards.

Pelvis:

Commonly referred to as the *hips*, the pelvis separates the lower body from the torso. Due to its location in the skeletal system, movement of the pelvis affects both the upper and lower body. From a skeletal perspective, the pelvis represents the base of the spine as well as the location where the femurs insert.

Pelvic Floor Muscles (PF):

Comprised of the *Levator Ani*, *Coccygeus* and connective tissue, the pelvic floor muscles essentially form a “hammock” that supports pelvic organs (i.e. bladder). The PF muscles assist in stabilization of the spine and pelvis.

Perceived Intensity Assessment (PIA):

Developed by UESCA, it is used to assess a client’s intensity threshold. The PIA utilizes the wall sit exercise to test one’s intensity threshold.

Periodization Cycles:

Represent a way to structure a training season. The periodization cycles are: *Macro*, *Meso* and *Micro*. The macrocycle represents the year. The mesocycle represents a block of time (denoted by a number of microcycles), typically between 2-6 microcycles. The microcycle is represented by a week.

Peroneals:

These muscles are on the lateral aspect of the lower leg. Made up of the *Peroneous Longus*, *Peroneous Brevis*. They control ankle plantar flexion and foot eversion.

Physical Activity Readiness Questionnaire (PAR-Q):

Used to determine if prospective clients are physically able and safe to begin an exercise program. Individuals that do not pass the PAR-Q cannot begin the exercise program without physician clearance.

Piriformis:

Abducts the femur when the hip is flexed. For many individuals, the sciatic nerve passes through the piriformis. Therefore when the piriformis tightens, the sciatic nerve becomes impinged and induces sciatica (pain that originates at the

gluteals and often travels down the leg via the sciatic nerve). This is called *Piriformis Syndrome*.

Plantar Fasciitis:

Plantar fascia is a fibrous band of fascia that runs from the heel to the forefoot. Irritation to the fascia often presents with heel or arch pain.

Plantar Flexion:

Increase angle between lower leg and foot. Women that wear high-heeled shoes are in a constant state of plantar flexion.

Plyometrics:

A power building exercise technique that is characterized by explosive movements.

Popliteus:

This is a small muscle located behind knee. The primary action is to flex the knee and more specifically, it is responsible for “unlocking” the knee from a straightened position. It also assists in internal rotation of the tibia. Over pronation of the foot or hyperextension of the knee during running can injure the popliteus. This injury often presents with difficulty and/ or pain when extending the knee.

Posterior:

Anatomical directional term referring to the back or toward the rear of the body.

Power:

The rate at which work is done. The equation for power is: $P = W/T$ (P = power, W = work and T = time).

Primary Mover:

A muscle or muscle group that is primarily responsible for produces a movement.

Pronation:

Medial rotation of a body part.

Prone:

Anatomical position in which the front surface is facing downward. Lying face-down is an example of the body in a prone position.

Proprioceptive Neuromuscular Facilitation (PNF):

A PNF stretch consists of a passive stretch followed by an isometric muscle contraction. During PNF stretching, a muscle is repeatedly contracted and relaxed, which triggers the golgi tendon reflex numerous times.

Protein:

A molecule composed of one or more chains of amino acids. Protein is necessary for the structure, function, and regulation of the body's cells, tissues, and organs.

Protraction:

Moves forward (anteriorly). Shoulders that are rounded to the front of the body are in a state of protraction.

Proximal:

Point of attachment, closer to the center of the body. The femur is proximal to the tibia.

Pubic Symphysis:

Connects the two sides of the lower pelvis (ischium). This joint is a semi-movable joint and is connected by ligaments and cartilage. The pubic symphysis expands as the distance between the legs expands. Pain in the pubic symphysis can occur when the joint does not return to the normal position (i.e. misaligned).

Pulmonary Circulation:

This system involves the heart and the lungs. Deoxygenated blood leaves the heart and goes to the lungs where it is re-oxygenated and returned to the heart.

Pyruvate:

An acid that is the result of glycolysis.

Q-Angle:

The angle between the two lines: Line 1-ASIS to Patella, Line 2-Tibial Tuberosity to Patella

Quadratus Lumborum:

Originates at the iliac crest and inserts into the lowest rib and side of lumbar vertebrae. Acts to stabilize the pelvis and to laterally flex the lumbar spine.

Quadriceps:

The “quads” as this muscle group is often called, is made up of four different muscles and are located on front of upper leg: *Vastus Lateralis*, *Vastus Medialis*, *Vastus Intermedius*, *Rectus Femoris*. These muscles are used to extend the lower leg and stabilize the knee. The rectus femoris also acts to flex the hip.

Radiculopathy:

Pain the extremities (arms, hand, feet) that originates from compression of spinal nerves.

Rate of Perceived Exertion (RPE) Scale:

A subjective test that measures physical exertion or pain.

R.I.C.E:

Acronym for **Rest-Ice-Compression-Elevation**. The purpose of R.I.C.E. is first-aid for soft tissue injuries and is used to minimize inflammation and increase comfort of an individual.

Recovery Heart Rate:

Assesses how fast the heart rate drops in a set duration of time after an intense effort. As an athlete becomes more aerobically conditioned, the faster the heart rate will recover.

Rectus Abdominis:

This muscle is made up of two muscles that run parallel to each other, and are separated by connective tissue (linea alba). The rectus abdominis is primarily responsible for flexion of the lumbar spine. Secondly, the rectus abdominis assists in stabilizing the lumbar spine via isometric contraction.

Rectus Femoris:

The rectus femoris is one of four muscles that make up the quadriceps and is visually the center muscle of the four when viewing the muscle in the frontal plane. The primary role of the rectus femoris is knee extension and hip flexion however it also plays a role in stabilizing the LPHC and the knee.

Reduced Muscle Activation:

Occurs when a muscle is overly tight. This causes the nervous system impulses to the overly tight muscle to decrease. This decrease affects the ability of the opposing (antagonist) muscle to contract properly.

Repeats:

These are characterized by running at a fast pace for a set amount of time (or distance), followed by a recovery period and then running the same distance and/ or pace again.

Resistance Training:

Any type of exercise that puts stress on a muscle(s) for the purpose of strength gains. Typically associated with calisthenics or lifting weights.

Resting Heart Rate:

Represents the heart rate at rest (before getting out of bed in the morning). This can be used as an assessment as typically the more “in-shape” one becomes, the lower the resting heart rate is.

Retraction:

Moves backward (posteriorly). Squeezing the shoulder blades together results in retraction of the shoulder blades.

Ribs:

Long, curved bones that protect the internal organs. Humans have 24 ribs (12 pairs)

Rhomboids:

The primary function of the rhomboids is to stabilize, rotate and retract the scapulas. The origin of the rhomboids is spine and they insert to the medial side of the scapulas.

Rotation:

Circular motion around a fixed point. Turning the torso in the transverse plane while in an upright, standing position is an example of rotation.

Rotator Cuff:

Muscles and associated tendons that control movement and stabilize the shoulder joint. Comprised of the following muscles and their respective actions:

- *Supraspinatus*: Shoulder Abduction
- *Infraspinatus*: External Shoulder Rotation
- *Teres Minor*: External Shoulder Rotation
- *Subscapularis*: Internal Shoulder Rotation

Sacroiliac Joint (SI Joint):

The SI Joint is the connection between the sacrum and the ilium, located to either side of the spine. The SI joint is stabilized by strong ligaments. The purpose of the joint is shock absorption of the spine. The joint locks when walking and running to provide a solid base of support during the foot push off aspect of the gait.

Sacrum:

The location where the lumbar spine connects to the coccyx. Made up of five fused vertebrae.

Sagittal Plane:

Divides the body into right and left halves. Walking in a straight line (forward or backwards) represents movement in the sagittal plane.

Sarcomere:

A unit within the myofibril that is responsible for muscle contraction. In essence, it's the functional unit of the muscle cell.

Sartorius:

This is the longest muscle in the body and begins (origin) below the ASIS and ends (insertion) on the superior and medial aspect of the tibia. This muscle acts to flex, abduct and externally rotate the hip. It is also aids in knee flexion.

Scapula:

Often referred to as the *shoulder blade*, it connects the humerus and clavicle.

Sciatica:

Pain down the posterior and or lateral aspect of the leg. This is more of a general term versus a specific condition. Sciatica can manifest itself as a result compression (i.e. herniated disc), spondylolisthesis or muscle contraction.

Scoliosis:

Spinal curvature in the frontal plane (lateral curvature). There are two classifications of scoliosis: *Structural* and *Functional*. The origin of functional scoliosis originates elsewhere in the body such as a leg length discrepancy whereas structural scoliosis originates due to a growth abnormality, in the spine. Structural scoliosis often presents with rotation or twisting of the spine as well as lateral curvature. Visible signs of scoliosis are uneven hips and shoulders.

Secondary mover (“stabilizer”):

A muscle or muscle group that contributes indirectly to produce movement. A secondary mover supports the primary mover. For example, during a push-up, the pectoralis major is the primary mover and the tricep is a secondary mover.

Selectorized Strength Machine:

A type of resistance exercise machine that typically focuses on one or two muscles. Characterized by a user being able to select the appropriate weight by inserting a “pin” into a stack of weights.

Serratus Anterior:

Originates at the ribs and attaches to the medial side of the scapula. Its function is to protract, stabilize and upward rotation of the scapula. This muscle is often termed the “boxer’s muscle” as it pulls the scapula forward, as occurs when throwing a punch.

Shear:

Force that is parallel to the surface.

Shin Spint:

A relatively generic term used to describe pain in the region of the tibia. Shin splints can be categorized into three main areas: Muscular, Tibial Fascia and Skeletal. If your client experiences shin pain, it is advised to have them seek out medical advice as the type of treatment will likely differ based on the “type” of shin spint.

Short Muscle:

Muscle is not elongated enough (i.e. tight) to provide adequate force production.

Shoulder Mobility Screen (SMS):

An assessment meant to assess the range of motion of the shoulder, primarily the muscles and connective tissue involved with internal and external rotation.

Sinoatrial Node (SA Node):

The heart's natural pacemaker. The SA Node sends an electric signal across the heart that causes a contraction.

Skeletal Muscle:

All muscles that are not cardiac or smooth. These are the muscles that provide movement of the body and account for 35-45% of a human's body weight (5). There are approximately 640 skeletal muscles in the body, most of which are grouped in pairs.

Skinfold Calipers:

An assessment tool use to pinch subcutaneous fat in various places on a subject's body for the purpose of determining their body composition.

Sliding Filament Model:

This model states that within the sarcomere, actin and myosin are alternately layered on top of one another (14). The myosin "ratchets" along the actin and causes the actin to shorten. This compression causes the sarcomere to shorten. Each sarcomere is connected to the next sarcomere by what is called a "Z line"—or Z-disk. When a muscle contracts, all of the sarcomeres contract at the same time. This simultaneous contraction produces force at each end of the muscle fiber.

Slow Twitch Muscle Fiber (ST):

Often referred to as "Type 1" muscle fibers. Characterized by the ability to contract repeatedly for long periods of time but with relatively low force. ST fibers rely on oxygen for energy and thus are red due to the many blood vessels.

Smooth Muscle:

Involuntary muscles that surround organs for the purpose of protection against outside forces

Soleus:

Deep muscle, located behind tibia and underneath the gastrocnemius. Acts to plantar flex and stabilize and ankle.

Spinal Stenosis:

Narrowing of the spinal canal as a result of degeneration, instability or a lesion. The *spinal canal* is a vertical hole within the vertebrae that allows the spinal cord to pass through.

Spin Bike:

Indoor bicycle that is characterized by a single speed and heavy flywheel. Often used in a group class format.

Spine:

Also referred to as the *Vertebral or Spinal Column*, the spine is made up of 33 vertebrae. It is divided up into five sections: Cervical, thoracic, lumbar, sacral and coccyx.

Spondylolisthesis:

A “slipped” vertebrae in relation to the one adjacent to it. Most commonly the vertebrae moves anteriorly however it can move laterally and posteriorly.

Static Stretch:

Stretch where the body part is held in a stretched, non-moving position.

Strength:

The ability of a person to product a force on an object using the musculoskeletal system.

Strength-Endurance Continuum:

States that strength and endurance are inversely related.

Strides:

Strides are essentially sprints (not 100% effort) that are done before and after runs or races. The purpose of doing them prior to a race is to warm up the legs to increase blood flow as well as to get the mind adapted to what race pace will feel like. Strides also work to enhance the neuromuscular efficiency of the runner.

Stroke Volume:

The amount of blood that is pumped from the heart (left ventricle) in one beat.

Subtalar Joint:

Connects the talus bone and calcaneous bone. This is a gliding joint and allows inversion and eversion of the ankle to occur. The *Achilles Tendon* covers the sub-talar joint when viewing the leg from a posterior aspect.

Superficial:

Directional term used to describe that a structure is closer to the surface of the body than another structure (i.e., Skin is superficial to bone)

Superior:

Used to describe an anatomical position where a landmark is above another landmark (toward the head). An example of this would be, the torso is superior to the legs.

Supination (ankle):

Also called “under pronation”, supination is essentially a lack of pronation.

Supine:

Anatomical position in which the front surface is facing upwards. Lying on the back is an example of the body being in a supine position.

Systemic Circulation:

This system relates to circulation throughout the body with the exception of the lungs. Oxygenated blood leaves the heart via arteries to supply the body and the de-oxygenated blood comes back to the heart via the veins.

Talocrural Joint:

Connects the tibia and fibula with the talus bone in the foot. This is a hinge joint and allows dorsi-flexion and plantar-flexion of the ankle.

Taper:

A systematic reduction of training volume and sometimes intensity. Done for the purpose of recovery and often done prior to an important event.

Tarsometatarsal:

This joint also spans across the width of the foot arch. Located posterior to the metatarsals, effectively at the junction between the midfoot and forefoot. This joint has more stability and less range of mobility than the midtarsal joint.

Tempo:

These workouts typically are performed at just below race pace and are done for 5-20 minutes (excluding warm up/cool down) at a time, depending on the length of the event being trained for. A substantial warm-up is recommended when performing a tempo efforts.

Tendinosis:

Damage to the structural integrity of a tendon.

Tendon:

Fibrous tissue that connects bone to muscle.

Tensor Fasciae Latae (TFL):

Acts to abduct the hip. The TFL also extends and externally rotates the hips. The TFL is located at the side of the hip.

Theraball:

Round inflatable ball that comes in a variety of sizes. Used for exercises.

Theraband:

Rubber-type band that is typically either flat and wide or in a tubing form and often has handles attached to either end. Used for resistance exercises.

Threshold Training:

The purpose of threshold training is to increase one's lactate threshold (LT). This training occurs in a range slightly below to slightly above one's MLSS. While threshold training can be steady efforts across a relatively long period of time, they can also be shorter or have variations of intensity. Tempo training is a type of threshold training.

Thoracic Spine:

Located between the cervical and lumbar section of spine. It is commonly referred to as the mid to upper back. The thoracic spine consists of 12 vertebrae (T1-12).

Tibia:

Bone in the lower leg that is adjacent to the fibula. Connects the femur and the ankle.

Tibialis Anterior:

Commonly referred to as the shin muscle. It is located on the anterior side of the lower leg (tibia). The action is dorsiflexion of the ankle.

Tibialis Posterior:

Located deep in the lower leg and posterior to the tibia, the tibialis posterior acts to provide support and stabilize the lower leg as well plantar flex the ankle.

Torque:

Rotational effect of force

Training Blocks:

Developed by UESCA, training blocks help to provide structure to an annual training program. There are five training blocks: *Base, Development, Peak, Taper, Off-Season*.

Training Log:

A recording of an athlete's training process. Most often the log is filled in daily and consists of a wide range of information including distance, physiological readings (heart rate) and energy level.

Transverse Abdominis (TVA):

This is the deepest of all the abdominal muscles and its primary action is to provide lumbar stability. The TVA acts as the body's natural weight belt by providing hoop tension around the midsection of an individual.

Transverse Plane:

Divides the body into top and bottom or more specifically, superior and inferior sections. The most common movement in the transverse plane is rotation. Rotating the upper body to one side is an example of movement in the transverse plane.

Trapezius:

A large diamond shaped muscle that consists of three parts: Upper, Middle and Lower. The primary action of the trapezius is to stabilize, elevate, retract and upwardly rotate the scapula.

Triceps Brachii:

Posterior muscle group of the upper arm (humerus). Extends the elbow. The triceps consists of three heads of the muscle: Lateral, medial and long.

Triglycerides:

Compounds that contain glycerol and free fatty acids. Glycerol is a chemical compound that is the primary chemical support structure of triglycerides (37). Glycerol and free fatty acids are the elements that the body catabolizes (breaks down) for energy. Glycerol is broken down anaerobically through glycolysis whereas fatty acids are catabolized through a process called beta-oxidation. During beta-oxidation, fatty acid is converted into Acetyl Co-A and then enters the Krebs Cycle to be oxidized.

Two-a-Days (TADS):

Two workouts in a single day.

Ultramarathon:

Any race over the marathon distance (>26.2 miles)

Unilateral:

Pertains to movement by one side of the body. Lifting the just one arm is an example of a unilateral movement.

USA Track and Field® (USATF):

The national governing body of the running and track & field in the United States

Valgus:

Inward angle of bone or joint (e.g., Knock Knees).

Varus:

Outward angle of bone or joint (e.g., Bow Legged).

Veins:

Carries deoxygenated (without oxygen) blood to the heart.

Ventilatory Threshold (VT):

The point at which the ventilation rate increases faster than the workload. Until the VT is reached, the workload and respiration rate increase linearly.

Vertebrae:

Bones that make up the spine.

VO₂ Max:

Often referred to as an individual's maximal oxygen consumption or as the test that assess one's aerobic capacity. The V stands for Ventilation and O₂ for oxygen. One's VO₂ Max refers to the amount of oxygen an individual can utilize in one minute and is expressed in ml O₂/kg-min. This means milliliters of oxygen per kilogram of body weight per minute.

Voluntary Muscle Contractions:

These are muscle contractions that are controlled by the central nervous system (CNS) and the origin of the contraction is a conscious thought by the brain. An example of this is a bicep curl.